Evaluation runs: rpc (rpc-at-most-once-aneris)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

No run of the stateless engine on this benchmark.

stateful

No run of the stateful engine on this benchmark.

offline

offline: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	14	conforms	0.2	8	10 min, 16 GB	0.2	0.0	0.0	in-memory	brain05, 2026-11-02
2	6	146	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	20	1070	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	70	6622	conforms	0.5	8	10 min, 16 GB	0.1	0.5	0.0	in-memory	brain05, 2026-11-02
5	252	37212	conforms	1.9	14	10 min, 16 GB	0.2	1.8	0.1	in-memory	brain05, 2026-11-02
6	924	196548	conforms	5.0	62	10 min, 16 GB	0.1	4.5	0.4	in-memory	brain05, 2026-11-02
7	3432	994422	conforms	21.3	291	10 min, 16 GB	0.3	19.4	1.9	in-memory	brain05, 2026-11-02
8	12870	4874870	conforms	107.6	1352	10 min, 16 GB	0.8	98.0	9.6	in-memory	brain05, 2026-11-02
9	48620	23327876	conforms	666.8	6994	10 min, 16 GB	1.8	614.9	51.9	in-memory	brain05, 2026-11-02

offline: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	14	conforms	0.2	8	10 min, 16 GB	0.2	0.2	0.0	in-memory	brain05, 2026-11-02
2	6	166	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	20	1350	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
4	70	9142	violation	0.6	8	10 min, 16 GB	0.1	0.5	0.0	in-memory	brain05, 2026-11-02
5	252	55692	violation	2.1	22	10 min, 16 GB	0.2	2.0	0.1	in-memory	brain05, 2026-11-02
6	924	316668	violation	6.5	110	10 min, 16 GB	0.1	5.8	0.7	in-memory	brain05, 2026-11-02
7	3432	1715142	violation	34.1	506	10 min, 16 GB	0.3	30.8	3.3	in-memory	brain05, 2026-11-02
8	12870	8958950	violation	205.9	2568	10 min, 16 GB	0.8	187.1	18.8	in-memory	brain05, 2026-11-02
9	48620	45498596	violation	1409.6	12761	10 min, 16 GB	1.8	1321.5	88.0	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): 12 dedup bag, 12 no-dedup bag.
skipped: a smaller size had reached its limit: 10 no-dedup bag, 10 dedup bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	14	conforms	3.2	168	10 min, 16 GB	0.2	3.1	0.0	in-memory	brain01, 2026-11-02
2	6	146	conforms	9.8	186	48 h, 80 GB	0.1	9.8	0.0	merge	pinky21, 2026-11-07

cross offline, Java implementation against the Rust specification: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	14	conforms	3.2	177	10 min, 16 GB	0.2	3.1	0.0	in-memory	brain01, 2026-11-02
2	6	166	violation	10.1	425	48 h, 80 GB	0.1	10.1	0.0	merge	pinky21, 2026-11-07

Not run (cross offline, Java implementation against the Rust specification). refused (an input missing at run time): 3 bag dedup, 3 bag no-dedup, 4 bag dedup, 4 bag no-dedup.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	14	conforms	2.9	167	10 min, 16 GB	2.9	0.0	0.0	in-memory	brain01, 2026-11-02
2	6	146	conforms	3.1	161	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
3	20	1070	conforms	3.1	173	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	6622	conforms	3.5	180	10 min, 16 GB	3.5	0.5	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	14	conforms	2.9	167	10 min, 16 GB	2.9	0.2	0.0	in-memory	brain01, 2026-11-02
2	6	166	violation	3.1	161	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3	20	1350	violation	3.1	173	10 min, 16 GB	3.1	0.2	0.0	in-memory	brain01, 2026-11-02
4	70	9142	violation	3.5	180	10 min, 16 GB	3.5	0.5	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 bag dedup, 5 bag no-dedup.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	14	conforms	2.9	163	1 h, 250 GB		brain01, 2026-11-02
2	6	146	conforms	3.1	157	1 h, 250 GB		brain01, 2026-11-02
3	20	1070	conforms	3.3	169	1 h, 250 GB		brain01, 2026-11-02
4	70	6622	conforms	4.2	176	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	14	conforms	2.9	163	1 h, 250 GB		brain01, 2026-11-02
2	6	166	violation	3.1	157	1 h, 250 GB	3	brain01, 2026-11-02
3	20	1350	violation	3.3	169	1 h, 250 GB	7	brain01, 2026-11-02
4	70	9142	violation	4.5	176	1 h, 250 GB	31	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	14	14	conforms	3.2	168	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
2	146	146	conforms	3.8	167	10 min, 16 GB	3.8	0.1	0.1	in-memory	brain01, 2026-11-02
3	1070	1070	conforms	8.0	786	10 min, 16 GB	7.9	0.1	0.0	in-memory	brain01, 2026-11-02
4	6622	6622	conforms	44.8	6399	10 min, 16 GB	44.8	0.5	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	14	14	conforms	3.2	177	10 min, 16 GB	3.1	0.2	0.0	in-memory	brain01, 2026-11-02
2	166	166	conforms	4.0	176	10 min, 16 GB	4.0	0.0	0.0	in-memory	brain01, 2026-11-02
3	1350	1350	conforms	8.5	787	10 min, 16 GB	8.5	0.2	0.0	in-memory	brain01, 2026-11-02
4	9142	9142	conforms	53.2	7013	10 min, 16 GB	53.2	0.5	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 5 bag dedup, 5 bag no-dedup.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	14	14	conforms	3.1	168	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
2	146	146	conforms	3.8	167	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain01, 2026-11-02
3	1070	1070	conforms	7.9	786	10 min, 16 GB	0.1	7.9	0.0	in-memory	brain01, 2026-11-02
4	6622	6622	conforms	44.8	6399	10 min, 16 GB	0.5	44.8	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	14	14	conforms	3.1	177	10 min, 16 GB	0.2	3.1	0.0	in-memory	brain01, 2026-11-02
2	166	166	conforms	4.0	176	10 min, 16 GB	0.0	4.0	0.0	in-memory	brain01, 2026-11-02
3	1350	1350	conforms	8.5	787	10 min, 16 GB	0.2	8.5	0.0	in-memory	brain01, 2026-11-02
4	9142	9142	conforms	53.2	7013	10 min, 16 GB	0.5	53.2	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 5 bag dedup, 5 bag no-dedup.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	14	14	conforms	3.3	161	10 min, 16 GB	3.3	0.8	0.0	in-memory	brain02, 2026-11-03
2	146	146	conforms	4.5	158	10 min, 16 GB	4.4	0.2	0.1	in-memory	brain02, 2026-11-03
3	1070	1070	conforms	8.7	758	10 min, 16 GB	8.5	0.7	0.2	in-memory	brain02, 2026-11-03
4	6622	6622	conforms	40.4	6738	10 min, 16 GB	39.7	0.9	0.6	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	14	14	conforms	3.3	164	10 min, 16 GB	3.3	0.4	0.0	in-memory	brain02, 2026-11-03
2	166	166	conforms	4.2	160	10 min, 16 GB	4.2	0.2	0.0	in-memory	brain02, 2026-11-03
3	1350	1350	conforms	8.7	813	10 min, 16 GB	8.7	0.6	0.0	in-memory	brain02, 2026-11-03
4	9142	9142	conforms	50.9	7312	10 min, 16 GB	50.9	0.7	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 5 bag dedup, 5 bag no-dedup.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	14	14	conforms	3.3	161	10 min, 16 GB	0.8	3.3	0.0	in-memory	brain02, 2026-11-03
2	146	146	conforms	4.5	158	10 min, 16 GB	0.2	4.4	0.1	in-memory	brain02, 2026-11-03
3	1070	1070	conforms	8.7	758	10 min, 16 GB	0.7	8.5	0.2	in-memory	brain02, 2026-11-03
4	6622	6622	conforms	40.4	6738	10 min, 16 GB	0.9	39.7	0.6	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	14	14	conforms	3.3	164	10 min, 16 GB	0.4	3.3	0.0	in-memory	brain02, 2026-11-03
2	166	166	conforms	4.2	160	10 min, 16 GB	0.2	4.2	0.0	in-memory	brain02, 2026-11-03
3	1350	1350	conforms	8.7	813	10 min, 16 GB	0.6	8.7	0.0	in-memory	brain02, 2026-11-03
4	9142	9142	conforms	50.9	7312	10 min, 16 GB	0.7	50.9	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 5 bag dedup, 5 bag no-dedup.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	14	14	conforms	4.6	167	1 h, 250 GB		brain02, 2026-11-03
2	146	146	conforms	4.5	167	1 h, 250 GB		brain02, 2026-11-03
3	1070	1070	conforms	10.4	778	1 h, 250 GB		brain02, 2026-11-03
4	6622	6622	conforms	42.0	6770	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	14	14	conforms	4.4	166	1 h, 250 GB		brain02, 2026-11-03
2	166	166	conforms	4.4	161	1 h, 250 GB		brain02, 2026-11-03
3	1350	1350	conforms	10.1	765	1 h, 250 GB		brain02, 2026-11-03
4	9142	9142	conforms	54.4	7249	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	14	14	conforms	4.3	157	1 h, 250 GB		brain02, 2026-11-03
2	146	146	conforms	4.4	154	1 h, 250 GB		brain02, 2026-11-03
3	1070	1070	conforms	8.7	754	1 h, 250 GB		brain02, 2026-11-03
4	6622	6622	conforms	40.6	6734	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-dedup**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	14	14	conforms	3.3	160	1 h, 250 GB		brain02, 2026-11-03
2	166	166	conforms	4.2	156	1 h, 250 GB		brain02, 2026-11-03
3	1350	1350	conforms	9.0	809	1 h, 250 GB		brain02, 2026-11-03
4	9142	9142	conforms	51.6	7308	1 h, 250 GB		brain02, 2026-11-03